INTRODUCTION

A school management system is a software application that is designed to streamline various administrative tasks within a school. This system is designed to automate a range of processes such as student enrolment, course scheduling, fee management, library management, attendance tracking, and more. The goal of a school management system is to simplify these processes and make them more efficient, saving time and reducing errors.

Key Features of a School Management System

Student Management

One of the primary functions of a school management system is to manage student information. This includes maintaining records of student enrolment, grades, attendance, and other key details. This information can be easily accessed and updated by teachers and administrators, reducing the time and effort required to keep track of student information.

Fee Management

Another important feature of a school management system is fee management. This includes the ability to track student fees, generate invoices, and receive payments. The system can also be configured to automatically generate reminders for overdue payments, making it easier for schools to manage their finances.

Course Scheduling

A school management system can also be used to schedule courses and manage class schedules. This feature allows teachers to easily plan and coordinate their schedules, reducing the risk of scheduling conflicts.

Attendance Tracking

Attendance tracking is an important part of any school management system. This feature allows schools to track student attendance in real-time, making it easier to monitor student engagement and identify students who may be at risk of falling behind.

- **Library Management**

A school management system can also be used to manage the school library. This includes the ability to track books and other resources, manage loans, and generate reports on the usage of the library.

- **Reporting and Analytics**

Finally, a school management system can provide powerful reporting and analytics capabilities. This includes the ability to generate reports on student performance, attendance, fees, and other key metrics. This information can be used to identify trends and make data- driven decisions to improve the quality of education.

Benefits of a School Management System

- **Improved Efficiency**

One of the biggest benefits of a school management system is improved efficiency. By automating a range of administrative tasks, schools can save time and reduce the risk of errors. This allows teachers and administrators to focus on more important tasks, such as improving the quality of education.

- **Better Communication**

Another benefit of a school management system is improved communication. This includes the ability to send automated notifications and updates to students and parents, reducing the need for manual communication.

- **Better Data Management**

A school management system can also improve the way that schools manage data. This includes the ability to store and access student information, grades, and other key data in a centralized location. This can make it easier for schools to monitor student progress and identify areas for improvement.

- **Improved Financial Management**

Finally, a school management system can improve the way that schools manage their finances. This includes the ability to track student fees, generate invoices, and receive payments. This can make it easier for schools to manage their budgets and reduce the risk of errors.

ABSTRACT

Student Management System is software which is helpful for students as well as the school authorities. In the current system all the activities are done manually. It is very time consuming and costly. Our Student Management System deals with the various activities related to the students. In the Software we can register as a user and user has of two types, student and administrator. Administrator has the power to add new user and can edit and delete a user. A student can register as user and can add edit and delete his profile. The administrator can add edit and delete marks for the student. All the users can see the marks.

School Management System helps headmasters to get the most accurate information to make more effective decisions. Teachers and headmasters gain time saving administrative tools, parents gain immediate access to their children's grades and students can track their own progress. School Management System equipped features makes it possible to generate schedules and reports in minutes and to retrieve attendance records, grade checks, report cards, transcripts, and form letters in just a few clicks. School Management Systems helps Teachers to complete grade book, track students attendance, input class notes, create lesson plans and detailed reports, and communicate with other staff members, students, and parents all via e-mail. It also helps Students to access assignments and tests, and view attendance records, grades, report cards, and progress reports all online. They also can communicate through mail and forums with teachers and other students online.

CODING

Sample Code -

```
import mysql.connector as mys  # importing mysql.connector module

# connecting with school database
conn = mys.connect(host="localhost", user="root", passwd="12345", database="school")
cur = conn.cursor()  # creating cursor object


# add_student() function will connect the student Admnno, Name, Grade, Address and
Contact particulars by input
def add_student():
    admnno = int(input("Enter Admission Number: "))
    sname = input("Enter student Name: ")
    grade = int(input("Enter grade: "))
    address = input("Enter address: ")
    contact = input("Enter contact number: ")
    sql = "INSERT INTO students (admnno, sname, grade, address, contact) VALUES (%s,
%s, %s, %s, %s)"
    cur.execute(sql, (admnno, sname, grade, address, contact))
    conn.commit()
    print("Student added successfully.")


# view_students() function will show the list of students with all particulars
def view_students():
    cur.execute("SELECT * FROM students")
    students = cur.fetchall()
    if students:
        print("\n--- Students List ---")
        for s in students:
            print(f"Admission No: {s[0]} | Name: {s[1]} | Grade: {s[2]} | Address: {s[3]} |
Contact: {s[4]}")
    else:
        print("No student records found.")
```

```
# display_student() function will display details of a particular student
def display_student():
    admnno = int(input("Enter Admission Number: "))
    sql = "SELECT * FROM students WHERE admnno = %s"
    cur.execute(sql, (admnno,))
    s = cur.fetchone()
    if s:
        print(f"\nAdmission No: {s[0]}\nName: {s[1]}\nGrade: {s[2]}\nAddress:
{s[3]}\nContact: {s[4]}")
    else:
        print("No student found with that admission number.")


# update_student() function will update the particulars of a student by input
def update_student():
    admnno = int(input("Enter Admission Number of the student to update: "))

    print("1. Update Name")
    print("2. Update Grade")
    print("3. Update Address")
    print("4. Update Contact")
    choice = input("Enter your choice: ")

    if choice == '1':
        value = input("Enter new name: ")
        sql = "UPDATE students SET sname = %s WHERE admnno = %s"
    elif choice == '2':
        value = int(input("Enter new grade: "))
        sql = "UPDATE students SET grade = %s WHERE admnno = %s"
    elif choice == '3':
        value = input("Enter new address: ")
        sql = "UPDATE students SET address = %s WHERE admnno = %s"
    elif choice == '4':
        value = input("Enter new contact number: ")
        sql = "UPDATE students SET contact = %s WHERE admnno = %s"
    else:
        print("Invalid choice.")
        return
```

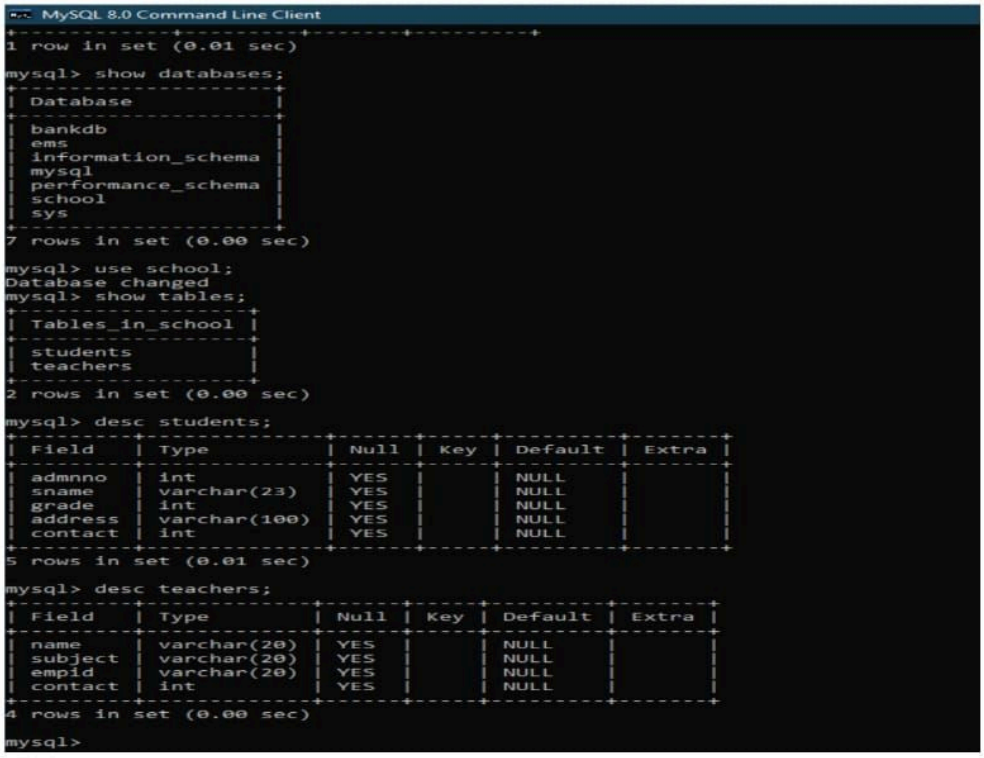
```
        print("4. Update Student")
        print("5. Search Student by Name")
        print("6. Delete Student")
        print("7. Exit")

        choice = input("Enter your choice (1-7): ")

        if choice == '1':
            add_student()
        elif choice == '2':
            view_students()
        elif choice == '3':
            display_student()
        elif choice == '4':
            update_student()
        elif choice == '5':
            search_student()
        elif choice == '6':
            delete_student()
        elif choice == '7':
            print("Goodbye!")
            break
        else:
            print("Invalid choice. Please try again.")

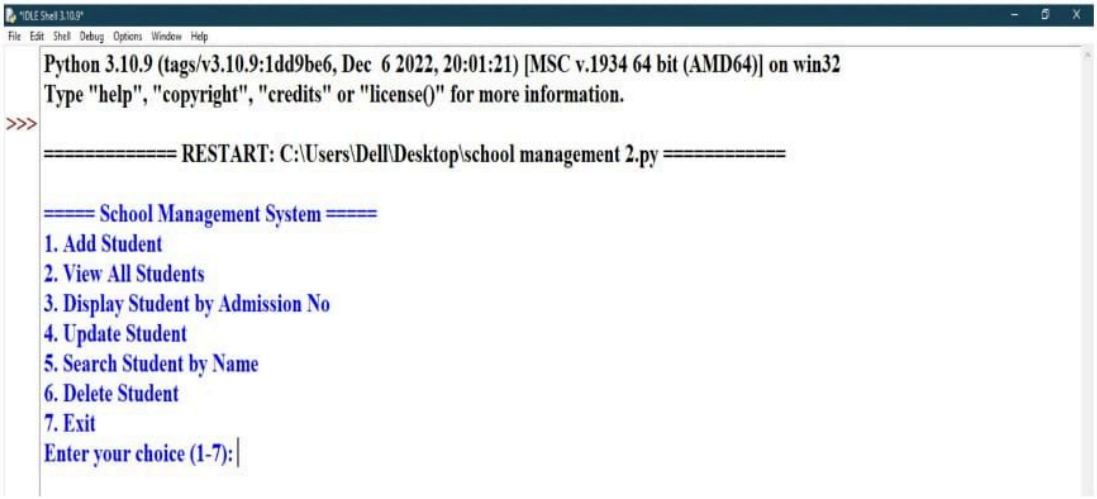

# run the program
if __name__ == "__main__":
    menu()
```

SQL TABLE CREATED IN SQL

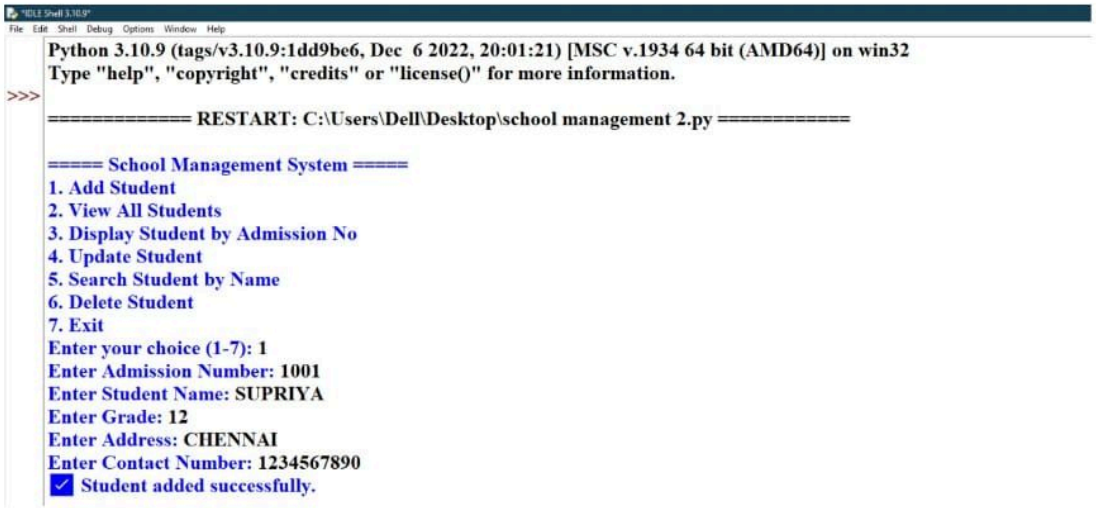


SNAPSHOTS

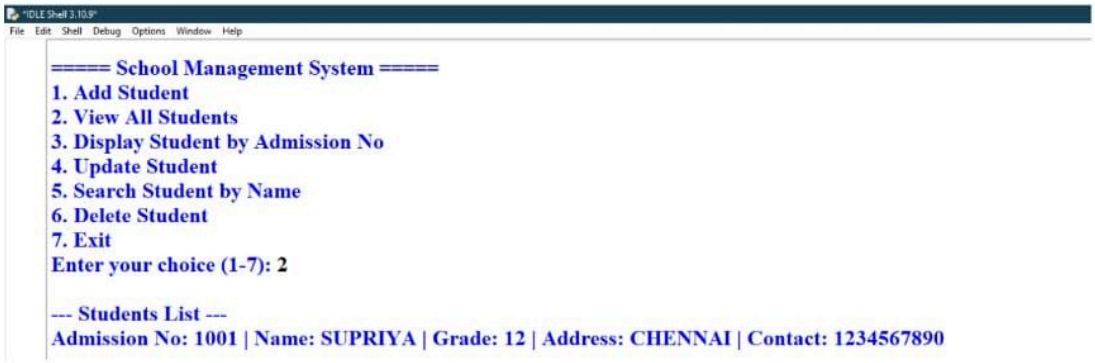
9.1 MENU



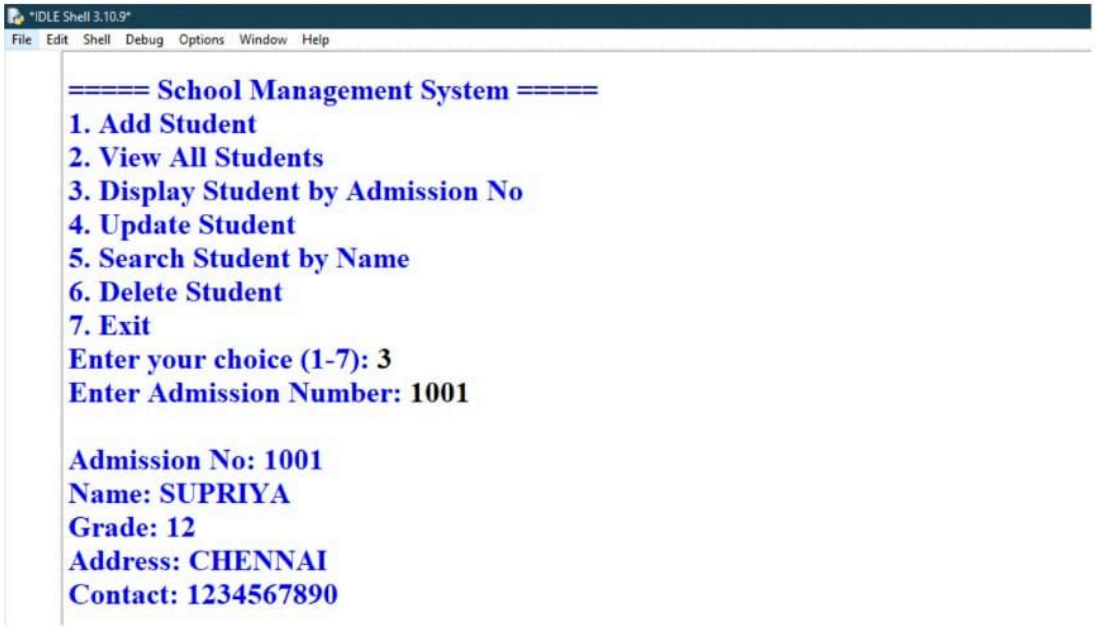
9.2 ADD STUDENTS



9.3 VIEW STUDENTS

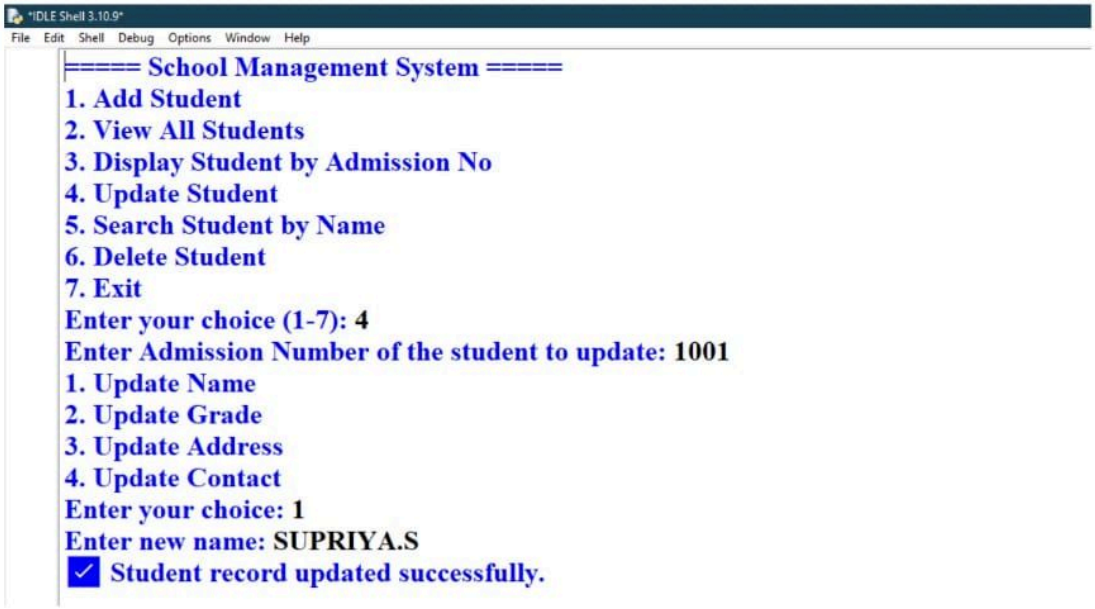


9.4 DISPLAY STUDENT

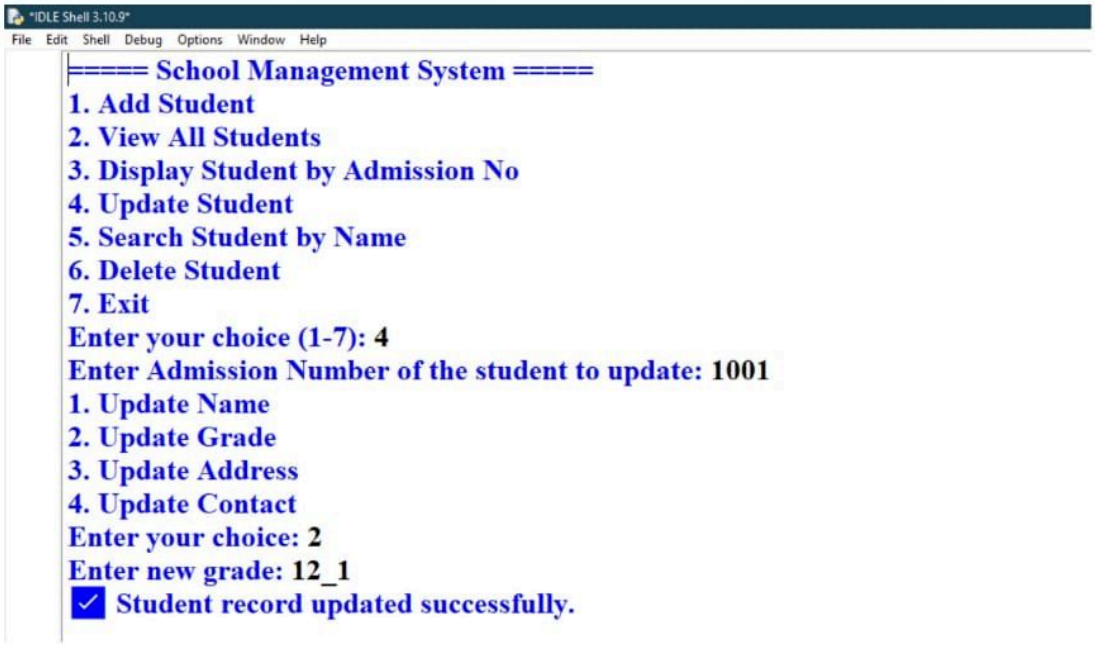


9.4 UPDATE STUDENT

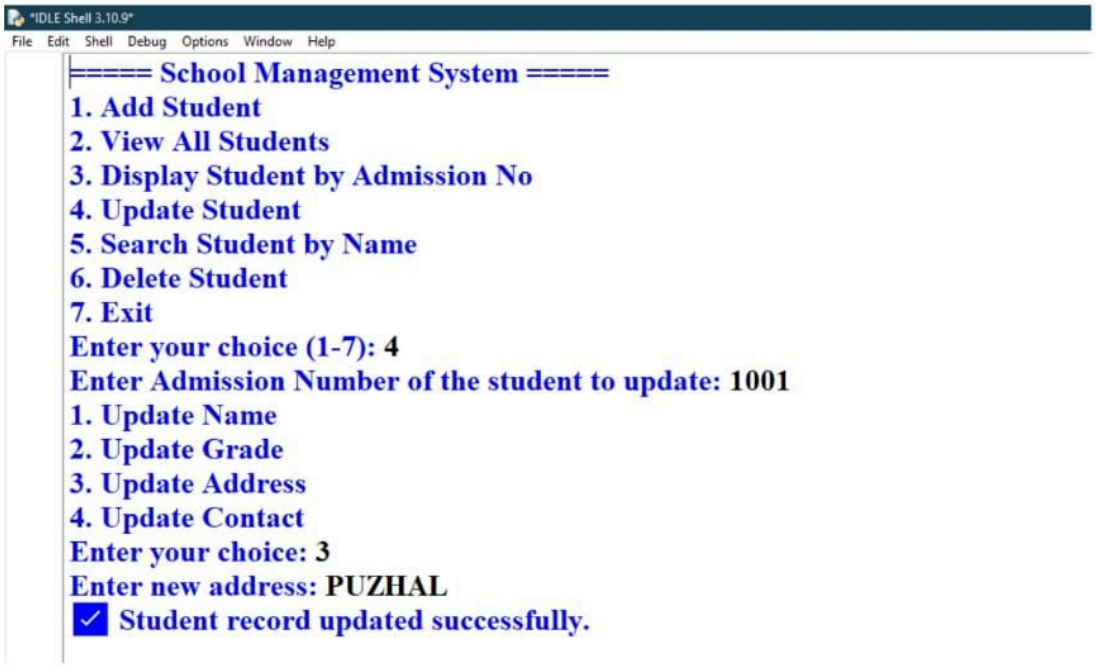
9.4.1 UPDATE STUDENT NAME



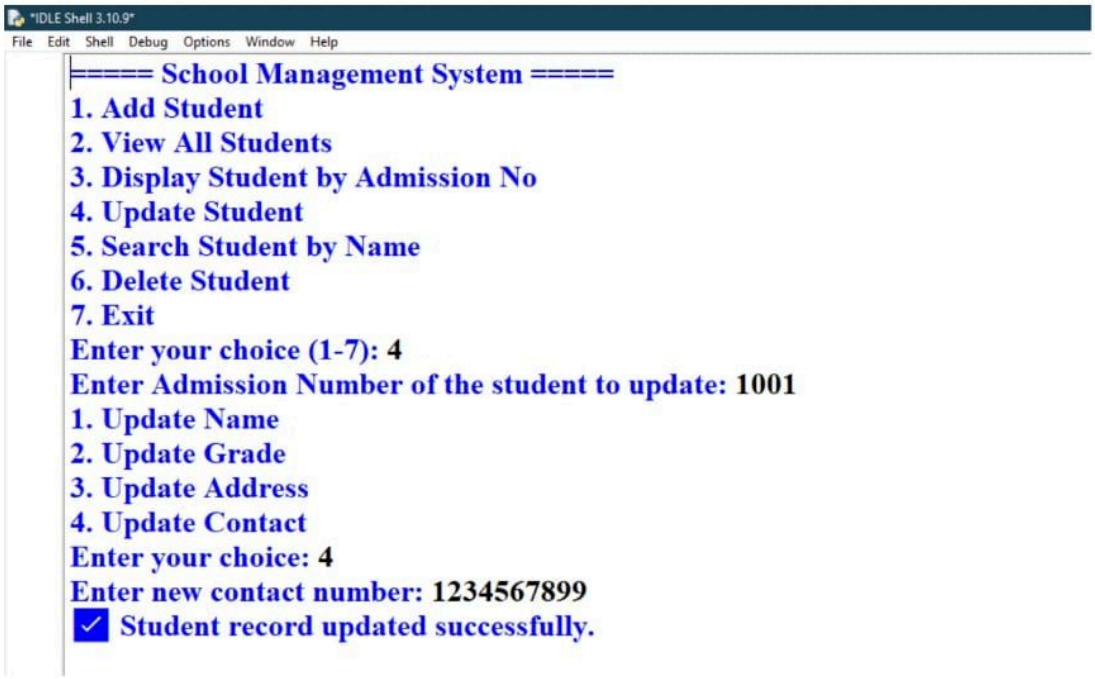
9.4.2 UPDATE STUDENT GRADE



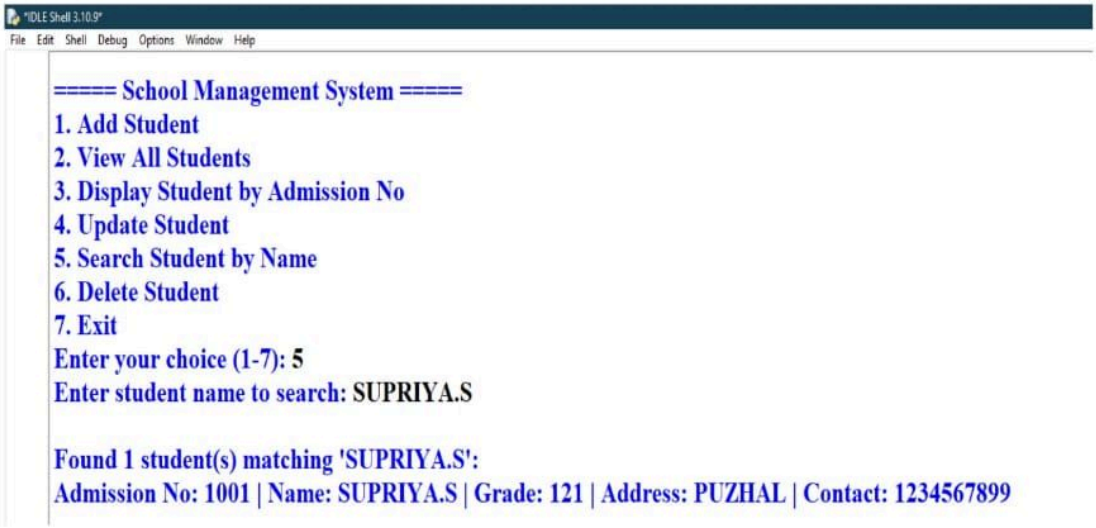
9.4.3 UPDATE STUDENT ADDRESS



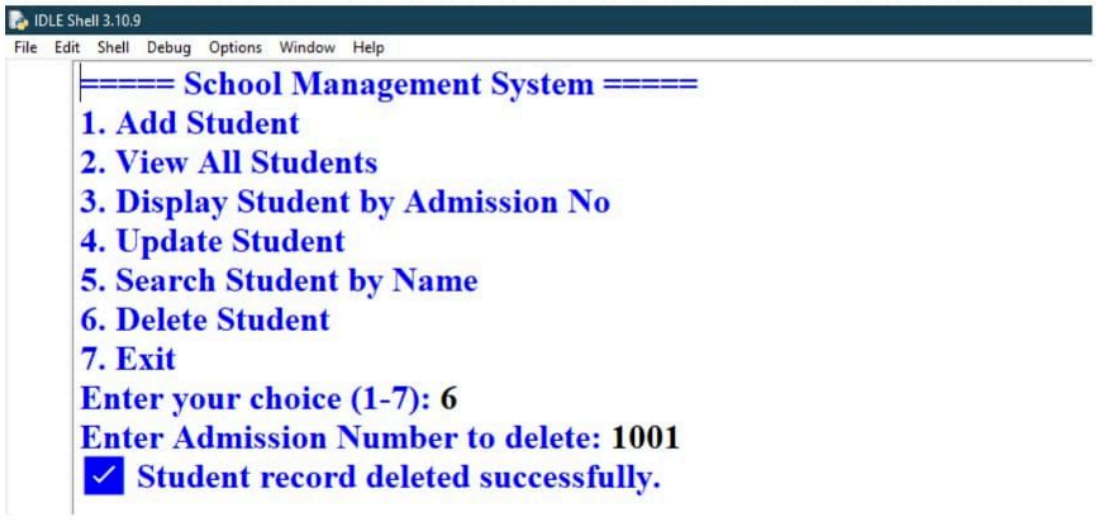
9.4.4 UPDATE STUDENT CONTACT



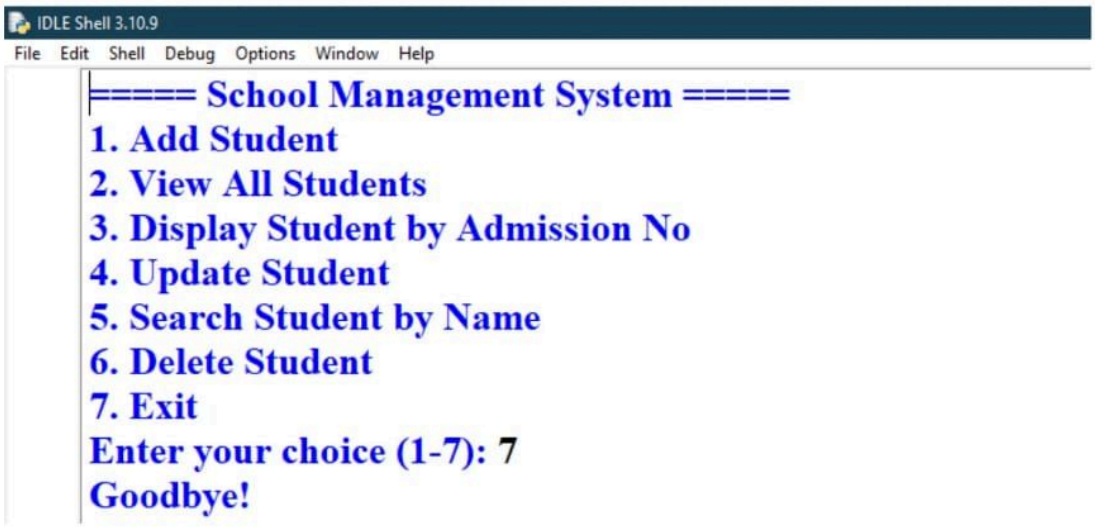
9.5 SEARCH STUDENT



9.6 DELETE STUDENT DETAIL



9.7 EXIT THE SYSTEM



BIBLIOGRAPHY

Reference Book

- Computer Science With Python [Sumita Arora] -Textbook For Grade Xii
- Python Pocket Reference Edition 5.0 – Mark Lutz
- Python In Nutshell -Edition 3.0 – Alex Martelli

WEBISTE

1. *https://www.python.org*
2. *https://www.w3schools.com*